

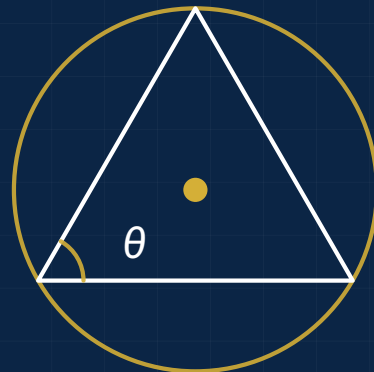


DR ESLAM AHMED
eliteigcse.com

CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 4

Circle Theorems

Eight facts, one habit: write the angle, then the theorem that gives it. Find radii (isosceles), diameters (90°) and tangents (90°) first.



5

VARIANTS

31

PAST-PAPER PARTS

52

PRACTICE QUESTIONS

YOU WILL MASTER

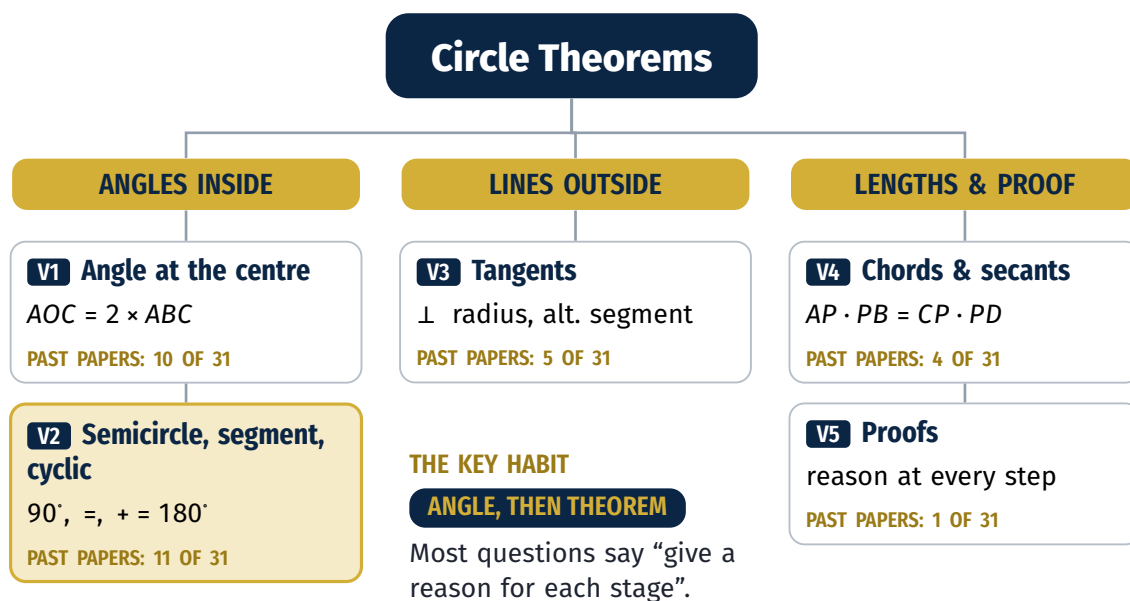
V1 Angle at the centre V2 Semicircle, same segment, cyclic

V3 Tangents V4 Chords & secants V5 Proofs with reasons

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Circle theorems at a glance



Your notes


★ How often does each variant come up?

24 past-paper questions (Jan 2020 → Nov 2025) hold 31 parts; 52 practice parts — grouped by type.

	PAST PAPERS (OUT OF 31)	PRACTICE PARTS
V1 Angle at the centre	10	7
V2 Semicircle, segment, cyclic	11	8
V3 Tangents	5	17
V4 Chords, secants	4	10
V5 Proofs with reasons	1	10

WHAT THIS TELLS YOU On every paper (Q11–Q20, 3–5 marks). Half the marks are for the reasons, written with the right words. Learn the eight sentences on the next page exactly.

BEFORE YOU START

The big idea

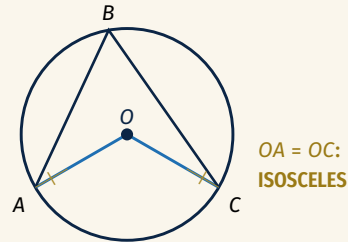


Your notes

Look for the hidden shapes first.

Two **radii** make an isosceles triangle. A **diameter** gives a 90° angle. A **tangent** meets the radius at 90° .

Mark these on the diagram, then the angle you need is usually one theorem away.


 The eight theorems — say them like this
Centre $2\times$

“angle at the centre is twice the angle at the circumference”

Semicircle 90°

“angle in a semicircle is 90° ”

Same segment

=

“angles in the same segment are equal”

Cyclic quad $+ = 180^\circ$

“opposite angles of a cyclic quadrilateral add to 180° ”

Tangent 90°

“a tangent is perpendicular to the radius”

Two tangents

=

“tangents from a point are equal in length”

Alternate segment

=

“angle between tangent and chord = angle in the alternate segment”

Chords

$$AP \cdot PB = CP \cdot PD$$

“intersecting chords”

**CALCULATOR FIRST, THEN CHECK**

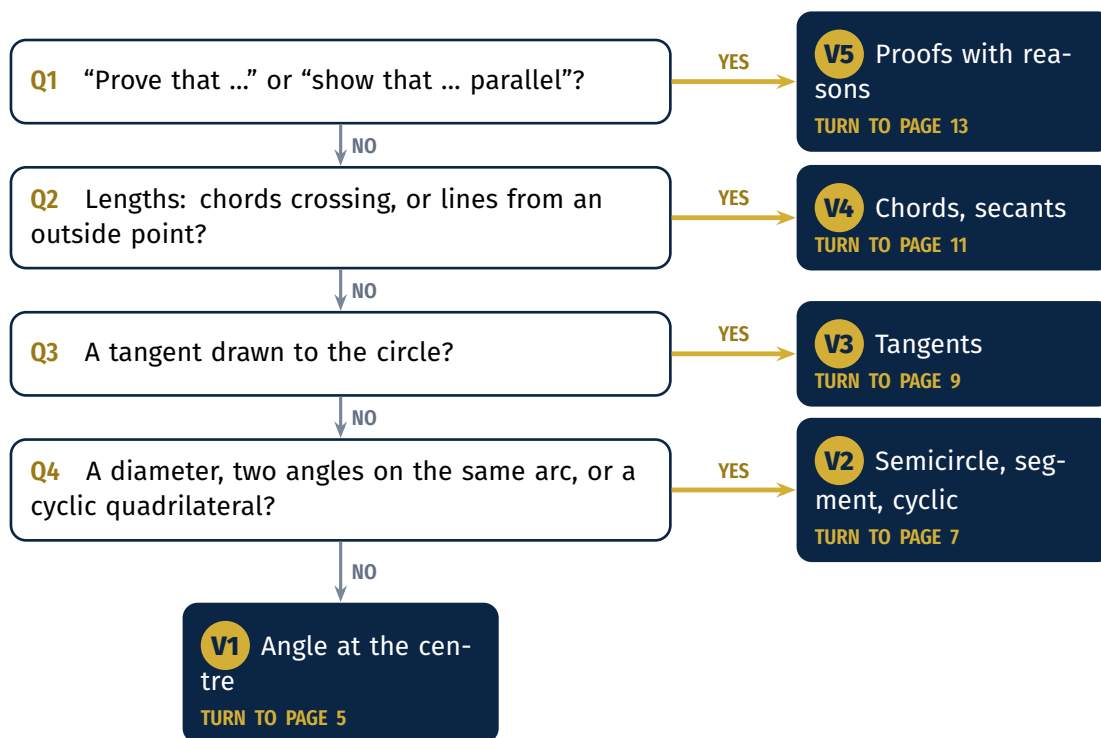
After you finish, check each triangle adds to 180° and each quadrilateral to 360° .

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

Angle at the Centre

The angle at the centre is twice the angle at the circumference on the same arc; radii make isosceles triangles.

SEEN IN PAST PAPERS

10 / 31

PRACTICE QUESTIONS: 7



Your notes

Came up in: Jan 2020 P1HR Q14 • Oct/Nov 2020 P1H Q16(b) • Jan 2021 P2H Q13 • Nov 2023 P1H Q14 • May 2024 P1HR Q13 • Nov 2024 P2H Q16 • Jun 2025 P2HR Q15 • Nov 2025 P2H Q15(a)

? What is it?

In plain words

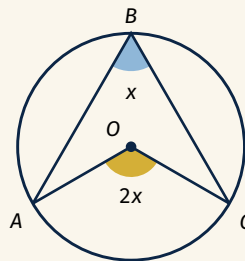
Lines from the ends of an arc meet at the **centre** and at the **circumference**. The centre angle is **double**.

OA, OB, OC are all radii: every triangle with two of them is **isosceles** — equal base angles.

IT LOOKS LIKE

AOC and ABC

reflex MOP



How do I do it?

1 Radii

Mark equal radii; write the isosceles base angles.

2 Same arc

Centre angle and circumference angle on the same arc.

3 Double / halve

Centre = $2 \times$ circumference.

4 Reason

Write the theorem's words for each angle.

CENTRE = $2 \times$ CIRCUMFERENCE

Same arc only.

V1 Angle at the Centre

VARIANT 1 • SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"A, B, C are points on a circle, centre O" → **Radii: isosceles triangles**

"Find angle AOC" with ABC given → **Double it**

"Find the size of the reflex angle" → **360- the obtuse centre angle**

**WORKED EXAMPLE** **PAST PAPER**

June 2025 • Paper 2HR • Q15 • 3 marks

A, B, C are on a circle, centre O. Angle $BAO = 36^\circ$, angle $BCO = 21^\circ$. Work out angle ACO.

STEP 1 $ABO = 36^\circ$, $CBO = 21^\circ$ ◀ $OA = OB$, $OB = OC$: isosceles

STEP 2 $ABC = 36 + 21 = 57^\circ$ ◀ add

STEP 3 $AOC = 2 \times 57 = 114^\circ$ ◀ centre = $2 \times$ circumference

STEP 4 $ACO = \frac{180-114}{2} = 33^\circ$ ◀ isosceles $OA = OC$

ANSWER 33° ✓ $114 + 33 + 33 = 180$ ✓**WATCH OUT**

✗ $AOC = \frac{1}{2}ABC$

✓ $AOC = 2 \times ABC$

The centre angle is the bigger one.

**EXAM TIP**

Write " $OA = OB$ (radii)" on the diagram — it unlocks the isosceles angles.

**Check yourself**

calculator first • answers on page 15

1.1

Nov 2023 P1H

A, B, C on a circle, centre O. Angle $OAC = 52^\circ$. Find angle ABC.

Answer:

1.2

Nov 2025 P2H

A, B, C, D on a circle, centre O. Angle $DAB = 48^\circ$. Find the obtuse angle DOB.

Answer:

1.3

Jan 2020 P1HR

A, B, C on a circle, centre O. Angle $ABC = 38^\circ$. Find angle OAC.

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Semicircle, Same Segment, Cyclic Quadrilateral

Diameter $\rightarrow 90^\circ$; same arc $\rightarrow$ equal angles;
cyclic quadrilateral $\rightarrow$ opposite angles add to 180° .

SEEN IN PAST PAPERS

11 / 31

PRACTICE QUESTIONS: 8


Your notes

Came up in: Jan 2020 P2H Q13 • Oct/Nov 2020 P1H Q16(a) • Jan 2021 P2H Q13(a)(ii) • May 2021 P2H Q18 • Jan 2022 P1HR Q14 • Jan 2022 P2H Q16 • Jun 2022 P1HR Q15 • Jun 2023 P2H Q14 • Nov 2024 P2H Q16(c) • Nov 2025 P2H Q15(b)

? What is it?

In plain words

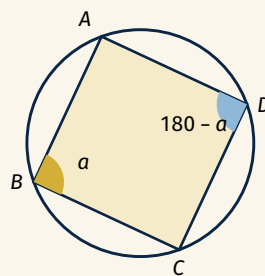
A triangle on a **diameter** has a right angle at the circumference.

Angles standing on the **same arc** (a “bow-tie”) are equal. A quadrilateral with all four corners on the circle: **opposite** angles add to 180° .

IT LOOKS LIKE

AOC diameter

ABCD cyclic



How do I do it?

1 Diameter?

Right angle at the circumference.

2 Same arc?

Bow-tie angles are equal.

3 Cyclic quad?Opposite angles: 180° .**4** FinishTriangle 180° for the last angle, with reasons.**DIAMETER = 90°** The most common variant.

V2 Semicircle, Same Segment, Cyclic Quadrilateral VARIANT 2 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“AOD is a diameter of the circle” → **Angle at the circumference = 90°**

Two angles from the same chord, on the same side → **Equal (same segment)**

Four points joined in order around the circle → **Opposite angles add to 180°**

WORKED EXAMPLE PAST PAPER

January 2020 • Paper 2H • Q13 • 4 marks

A, B, C, D on a circle, centre O; AOD is a diameter. Angle $CBD = 28^\circ$, angle $BDA = 32^\circ$. Find angle BDC.

STEP 1 $ABD = 90^\circ$ ◀ angle in a semicircle

STEP 2 $BAD = 180 - 90 - 32 = 58^\circ$ ◀ angles in a triangle

STEP 3 $BCD = 180 - 58 = 122^\circ$ ◀ cyclic quadrilateral ABCD

STEP 4 $BDC = 180 - 122 - 28 = 30^\circ$ ◀ angles in triangle BCD

ANSWER 30°

✓ $122 + 28 + 30 = 180$ ✓

WATCH OUT

✗ opposite angles are equal
✓ they add to 180°

That is a parallelogram, not a cyclic quadrilateral.

EXAM TIP

Name each angle with three letters — “angle B” scores nothing when four lines meet there.

Check yourself

calculator first • answers on page 15

2.1 Jan 2022 P2H

EOG is a diameter; angle $EGD = 42^\circ$; F is on the circle. Find angle DFG.

Answer:

2.2 Jun 2022 P1HR

PS is a diameter; PQRS cyclic; angle $PQR = 136^\circ$. Find angle RPS.

Answer:

2.3 Dr Eslam

ABCD is a cyclic quadrilateral with angle $A = 73^\circ$. Find angle C.

Answer:

V3

VARIANT 3

Tangents

Tangent $\perp$ radius; two tangents from a point are equal; tangent–chord angle = angle in the alternate segment.

SEEN IN PAST PAPERS

5 / 31

PRACTICE QUESTIONS: 17



Your notes

Came up in: Jan 2021 P1HR Q14 • May 2021 P1H Q15 • Nov 2021 P2H Q11 • Jun 2023 P2HR Q20 • Jun 2023 P2H Q7

? What is it?

In plain words

A tangent touches the circle once and meets the **radius at 90°** .

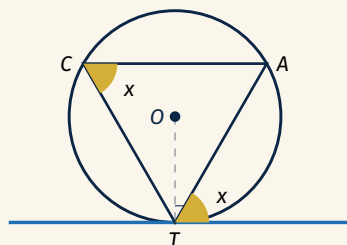
The angle between a tangent and a chord equals the angle the chord makes **on the other side** of the circle (**alternate segment**).

IT LOOKS LIKE

TRV tangent at R

tangents DA

DC



How do I do it?

1 Right angles

Radius to the tangent point: 90° .

2 Two tangents

Equal lengths: kite, isosceles.

3 Alternate segment

Tangent–chord angle $\rightarrow$ angle across the circle.

4 Finish

Triangle / quadrilateral sums, with reasons.

TANGENT $\perp$ RADIUS Mark the 90° first.

V3 Tangents

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“*TRV is the tangent to the circle at R*” → $ORV = 90^\circ$; **alternate segment**

“*DA and DC are tangents*” → $OAD = OCD = 90^\circ$; **quadrilateral 360°**

“*AB = CB*” with a tangent at C → **Alternate segment + isosceles**

**WORKED EXAMPLE** **PAST PAPER**

May 2021 · Paper 1H · Q15 · 4 marks

P, Q, R on a circle, centre O; TRV is the tangent at R. Reflex angle POR = 238°, angle QRV = 60°. Find angle OPQ.

STEP 1 $ORQ = 90 - 60 = 30^\circ$ ◀ tangent $\perp$ radius

STEP 2 $OQR = 30^\circ$ ◀ isosceles, $OQ = OR$

STEP 3 $PQR = \frac{1}{2}(360 - 238) = 61^\circ$ ◀ centre = $2 \times$ circumference

STEP 4 $OQP = 61 - 30 = 31^\circ = OPQ$ ◀ isosceles, $OP = OQ$

ANSWER 31° ✓ triangle OPQ: $31 + 31 + 118 = 180$ ✓**WATCH OUT**

✗ $PQR = \frac{238}{2}$

✓ $\frac{122}{2} = 61^\circ$

Q stands on the arc *PR* that faces the obtuse angle 122° .

**EXAM TIP**

Write “alternate segment theorem” in full — “alternate angles” is a different fact.

**Check yourself**

calculator first · answers on page 15

3.1

Jun 2023 P2H

DA and DC are tangents at A and C; angle $ADC = 48^\circ$. Find reflex angle AOC.

Answer:

3.2

Jun 2023 P2HR

$AB = CB$; MN is the tangent at C; angle $BCN = 78^\circ$. Find angle ABC.

Answer:

3.3

Dr Eslam

Tangents from P touch at A and B; angle $APB = 40^\circ$. Find angle AOB.

Answer:

V4

VARIANT 4

Chords & Secants

Crossing chords: $AP \cdot PB = CP \cdot PD$. From an outside point: whole $\times$ outside part is equal.

SEEN IN PAST PAPERS

4 / 31

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Jan 2020 P2HR Q18 • Jan 2022 P2H Q19 • Jun 2023 P2HR Q18 • Jun 2025 P1HR Q23

? What is it?**In plain words**

Two chords crossing at P : the two pieces of one chord **multiply** to the same as the two pieces of the other.

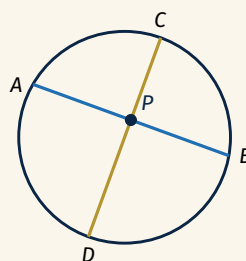
Two lines from a point P **outside**: (outside part) $\times$ (**whole** line) is the same for both.

IT LOOKS LIKE

$AP = 9$

$PB = 6$

$CP = 8; PB \cdot PA = PC \cdot PD$



$AP \times PB = CP \times PD$

How do I do it?**1 Inside or outside?**

Chords cross inside, or lines meet outside.

2 Pieces

Inside: the two parts. Outside: outside part and whole.

3 Equation

Products equal; solve.

4 Answer

The length asked (a piece, a whole, or the radius).

PIECE $\times$ PIECE = PIECE $\times$ PIECE

Outside: whole, not the chord.

V4 Chords & Secants

VARIANT 4 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"APB and CPD are chords" → $AP \cdot PB = CP \cdot PD$

"ABP and DCP are straight lines" (P outside) → $PB \cdot PA = PC \cdot PD$

"PTQ is a diameter ... find the radius" → **Find the missing piece, then halve**



WORKED EXAMPLE PAST PAPER

June 2023 • Paper 2HR • Q18 • 3 marks

ABP and DCP are straight lines from P outside the circle. AB = 16 cm, BP = 14 cm, CP = 12 cm. Work out the length of DC.

STEP 1 $PA = 16 + 14 = 30$ ◀ whole line

STEP 2 $PB \times PA = PC \times PD$ ◀ secant rule

STEP 3 $14 \times 30 = 12 \times PD \Rightarrow PD = 35$ ◀ solve

STEP 4 $DC = 35 - 12 = 23$ ◀ the chord

ANSWER 23 cm

✓ $12 \times 35 = 420 = 14 \times 30$ ✓



WATCH OUT

✗ $14 \times 16 = 12 \times DC$

✓ $14 \times 30 = 12 \times PD$

Outside: multiply by the whole line from P.



EXAM TIP

Label every length on the diagram before writing the equation.



Check yourself

calculator first • answers on page 15

4.1

Jan 2020 P2HR

Chords APB, CPD:
AP = 9, PB = 6, CP = 8.
Find PD.

Answer:

4.2

Jan 2022 P2H

PTQ is a diameter, RTS
a chord: TQ = 3, ST = 4,
TR = 12. Find the
radius.

Answer:

4.3

Dr Eslam

Chords AB, CD cross at
P: AP = 6, PB = 4,
CP = 3. Find PD.

Answer:

V5

VARIANT 5

Proofs with Reasons

A chain of angles, each with its theorem,
ending in the fact you were asked to prove.

SEEN IN PAST PAPERS

1 / 31

PRACTICE QUESTIONS: 10



Your notes

Came up in: Oct/Nov 2020 P1HR Q15

? What is it?**In plain words**

“Prove” means every step needs a **reason**. Work towards what you must show: two **equal alternate angles** prove lines are **parallel**; two equal angles pairs prove triangles **similar**. Finish with a sentence that says what is proved.

IT LOOKS LIKEprove $ABC \parallel EF$

show similar

 $FDB = 70^\circ$ alt. segment

 $EDB = 40 + 70 = 110^\circ$
 $EFB = 70^\circ$ cyclic quad

 $EFB = FBC$: alternate, so parallel
How do I do it?**1 Target**

What fact proves it? (alternate angles, two angle pairs ...)

2 Chain

Find angles towards it, one theorem each.

3 Reasons

Write the theorem after every angle.

4 ConcludeOne sentence: “so ABC is parallel to EF ”.**EVERY LINE HAS A REASON** And a final sentence.

V5 Proofs with Reasons

VARIANT 5 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Prove that the tangent ABC is parallel to EF ” → **Make alternate angles equal**

“Prove that triangles ... are similar” → **Two pairs of equal angles**

“Give a reason for each stage” → **The theorem in words after each line**

**WORKED EXAMPLE****PAST PAPER**

Oct/Nov 2020 • Paper 1HR • Q15 • 4 marks

B, D, E, F on a circle; ABC is the tangent at B . Angle $EDF = 40^\circ$, angle $FBC = 70^\circ$. Prove that ABC is parallel to EF .

STEP 1 $FDB = FBC = 70^\circ$ ◀ alternate segment theorem

STEP 2 $EDB = 40 + 70 = 110^\circ$ ◀ add

STEP 3 $EFB = 180 - 110 = 70^\circ$ ◀ opposite angles of cyclic quadrilateral $EDBF$

STEP 4 $EFB = FBC = 70^\circ$ ◀ alternate angles are equal

ANSWER so ABC is parallel to EF

✓ each step has a reason ✓

**WATCH OUT**

✗ “they look parallel”

✓ “alternate angles $EFB = FBC$ ”

A proof needs a fact, not the picture.

**EXAM TIP**

Underline your final sentence — it is the last mark.

**Check yourself**

calculator first • answers on page 15

5.1

Practice • Medium

A tangent at A makes 65° with chord AB . C is on the major arc. Find angle ACB .

Answer:

5.2

Jan 2021 P1HR

ABC is the tangent at B ; angle $ABD = 39^\circ$, angle $EFD = 18^\circ$; $BDEF$ cyclic. Find angle BDE .

Answer:

5.3

Dr Eslam

Chords AC and BD cross. Angle $BAC = 31^\circ$. Find angle BDC and give the reason.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 38° **1.2** 96° **1.3** 52° **2.1** 48° **2.2** 46° **2.3** 107° **3.1** 228° **3.2** 24° **3.3** 140° **4.1** 6.75 cm**4.2** 9.5 cm**4.3** 8**5.1** 65° **5.2** 123° **5.3** 31°

THE WHOLE TOPIC ON ONE PAGE

Circle theorems cheat sheet



Your notes
V1 Centre 10/31

$$AOC = 2 \times ABC$$

Radii: isosceles.

V2 Semicircle etc. 11/31

$$90^\circ, =, 180^\circ$$

Diameter; same segment; cyclic.

V3 Tangents 5/31 $\perp$ radius

Two tangents equal; alternate segment.

V4 Chords 4/31

$$AP \cdot PB = CP \cdot PD$$

Outside: part $\times$ whole.**V5 Proofs 1/31**

alternate angles

Reason every line; conclude.

**TOP 3 LOST MARKS**

1. reasons missing or vague $\times$
2. halving instead of doubling $\times$
3. outside secant: chord, not whole $\times$

BOTTOM LINE Radii, diameters, tangents first $\cdot$ angle, then theorem $\cdot$ centre
 $= 2 \times \cdot$ cyclic $+ = 180$.